

Real-Time Vibration Estimation and Compensation With Long Short-Term Memory Recurrent Neural Network

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Abstract—Vehicles, as the moving platforms of various activities, have played important roles in modern society. However, the mechanical vibration due to various sources greatly degrades the performance of on-board devices that require high precision. To compensate the vibration, the technical challenges include: 1) the vibration possesses multiple time-varying dominant frequencies; 2) the broad bandwidth; 3) the phase difference between compensating movement and vibration; and 4) realizing real-time (RT) operation. In this article, we propose an AI-aided RT estimation and compensation method to address these challenges. The proposed method consists of two recursive least square-based filters to remove the gyroscope noise and drift, and a long short-term memory-based recurrent neural network to remove the phase shift. Applied techniques are all implemented in RT. The method is validated by simulations and RT experiments using vibration data sampled from a real vehicle and achieves a 75% compensation rate, which outperforms existing methods.

Index Terms—Adaptive filter, long short-term memory (LSTM), real-time (RT) vibration compensation, recursive neural network (RNN), signal processing, time series forecasting.

I. INTRODUCTION

VEHICLES, including but not limited to automobiles, ships, and aircraft, are important platforms for various industrial and scientific activities. However, the on-board vibration is a critical disturbance to various precision devices. Taking the vehicle-based free space optical communication which is of high interest in recent years [1], [2] for example, as shown Fig. 1, a tiny angular vibration of $10\mu\text{rad}$ could be amplified to a 1-m error in a 10-km task, causing communication failure.

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Traditional vibration control like isolation platforms and suspension systems are mostly passive methods, which possess residue and hence are insufficient in long-distance applications. Active compensation is hence of interest. This approach includes two stages: 1) estimation and 2) actuation. The encountered vibration is first sensed and estimated to signal. Next, based on the obtained signal, the affected devices or optical paths are actuated to the opposite direction of the vibration to maintain stability.

To realize active compensation, the technical challenges include 1) the multiple time-varying dominant frequencies in vibration; 2) broad signal bandwidth up to 100 Hz; 3) the phase difference between compensating movement and vibration; and 4) the operation should be in real-time (RT) to be effective.

As shown in Fig. 1, a vehicle platform includes various potential sources that may induce vibration of distinctive frequencies [3]. Resultant vibration hence possesses multiple dominant frequencies in a broad bandwidth. From observation and study, the bandwidth can be up to 100 Hz [4]. Moreover, most sources are dynamic systems, causing the frequencies to vary over time.

As to the phase difference, there are two sources: 1) the system processing time and 2) the applied filters. The mainstream RT systems typically have a few milliseconds delay which is negligible for slow RT applications, but not for signals up to 100 Hz. Specifically, a 1-ms delay will induce a phase shift as large as 10% of the period, resulting in significant error. The other source is the conventional filters. In vibration sensing and estimation, sensor selection is critical. Interferencing sensors, like the ultrawideband (UWB) devices, require sensor networks to be set [5], [6], making it impractical; while external referencing sensors like cameras have various problems in vibrating conditions [7], limiting the usage. Self-referencing sensors, conversely, are widely accepted in such cases. However, the sensors possess noise and drift problems. Utilizing filters to remove these disturbances is necessary, but conventional filters will cause a frequency-dependent phase shift which can hardly be specified and removed.

To address all challenges in vibration estimation, one may consider single-step or multiple-step approaches. A single-step method focuses on constructing an analytical model of the vibration with the sensor input. As the model converges, the sensor disturbances are naturally removed. The phase can also be corrected by model manipulation. Sharing this idea, the

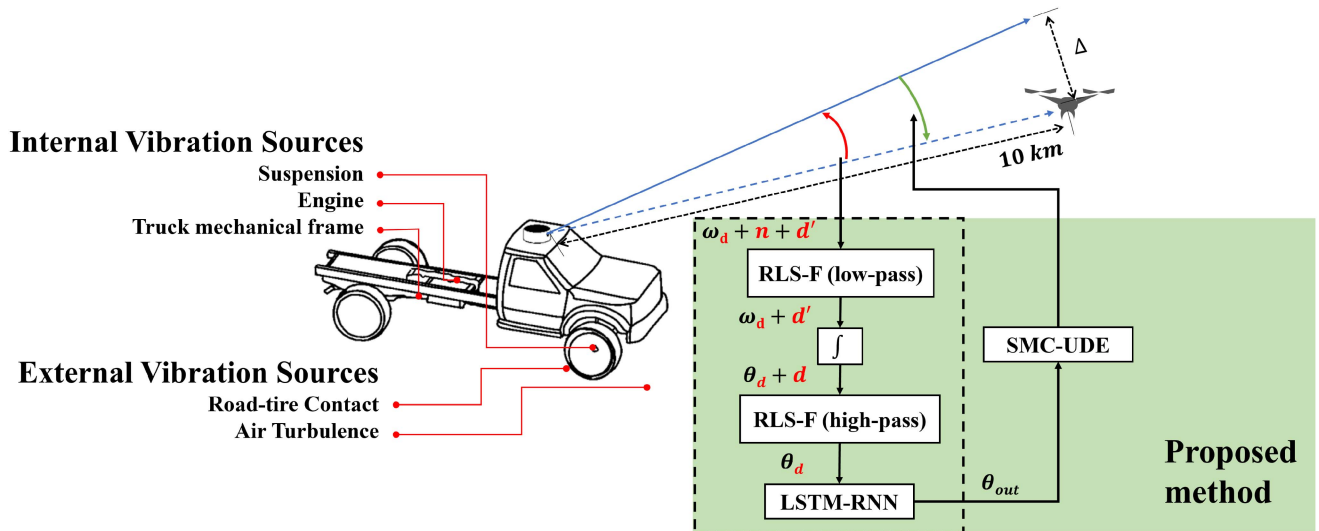


Fig. 1. Typical vehicle-based laser emission task and the proposed compensation approach. The sources of vibration and technical challenges are shown in red. The proposed method includes estimation and actuation stages. As the main contribution of this work, the estimation stage shown in the dotted square includes 2 RLS-F, and LSTM-RNN to remove the noise, drift, and phase shift to achieve RT compensation.

Fourier linear combiners (FLC)-based methods [8], [9] are widely used for signal estimation; while the dynamic harmonics regression method works for delay correction [10]. The methods are well tested in various fields, such as network delay compensation in telerobotics [11], hand tremor estimation [12], etc. However, all mentioned works focus on signals of narrow bandwidth. The multiple dominant frequencies, time variance, and broad bandwidth of vehicle vibration can lead to difficulties in model construction and the local minimum problem in model convergence.

With the rapid development of AI, neural networks (NN)-aided vibration control methods are of high interest in recent years. Shang et al. [13] proposed an NN-based controller for vibration suppression in two-inertia systems. In [14], an adaptive NN band-pass filter (ANNBPF)-based rotor vibration control method was proposed. In literature, the vibration to be suppressed in those systems differs greatly from the vehicle-based vibration in features. Moreover, the control strategies are not active compensation either. Applying those methods is not feasible.

We hence look into multiple-step approaches dealing with the delay-induced phase shift and sensor disturbances separately in estimation. The delay of an RT system is typically constant, which can be compensated by forecasting. Conversely, phase shifts caused by the conventional filters in sensor disturbances removal can hardly be determined. Alternative filtering methods are of interest.

The most classic time series forecasting methods include exponential smoothing and the autoregressive integrated moving average (ARIMA) [15], with well proven performance in decades. However, for periodic time series like the vibration signals, the forecasting is based on historical periods and is limited in addressing the time variance between periods. Decomposition methods can better address the complex period features [16] but with cost in huge data demand making the RT usage challenging.

Recently, AI-based techniques are also proposed with improved accuracy in time series processing [17]. Traditional NN considers static and single predictions, which does not address time-varying features. Therefore, recurrent NNs (RNN) are introduced for dynamic and sequential tasks [18] but with the vanishing gradient problem. The long short-term memory (LSTM) unit is hence developed [19] and is widely used in offline forecasting of pricing [20], air condition [21], energy consumption [22], etc. In [23], [24], and [25], RT or near RT forecasting with LSTM networks is proposed for temperature, waiting time, and disturbance detection. However, the involved time series or tasks possess relatively simple features which are mainly of a trend or single dominant frequency. With less computation than the LSTM, gated recurrent unit (GRU) is developed. Ou et al. [26] proposed a GRU network-based control strategy for planar motors. The network predicts the trajectory modification of the motion with complex frequency features for precision control. However, the network works before the motion is performed, which is not in RT. Moreover, GRU networks have only one state vector, which may slightly degrade the performance.

As to sensor noise and drift filtering, AI-aided adaptive filters are also proposed [27], [28]. The idea is to map the conventionally filtered phase-distorted signal to the offline filter results. However, as the phase shift is frequency-dependent, the generalization of these methods to signals with broader bandwidth lacks further discussion.

To this end, we propose an RT estimation approach for vehicle vibration, addressing the aforementioned challenges. As shown in Fig. 1, the approach works in a multistep manner, which consists of recursive least square filters (RLS-F) to remove the sensor disturbances and a novel forecasting LSTM-RNN to eliminate the delay-induced phase shift. The RLS-F is based on regression models to the input signal, which is an adaptive approach with negligible phase shift compared to conventional

filters. As to the LSTM-RNN, the design takes the idea of ARIMA which uses the most recent historical data as input and forecasts a future value of the vibration series. As the mainstream time series processing network, LSTM possesses well-tested performance in processing series with complex features. Moreover, compared to the cutting edge Transformer networks, the LSTM network typically requires much fewer layers for satisfactory performance, facilitating RT application. The proposed design is capable of forecasting vibration signals with the aforementioned complex features and works in RT, which can hardly be addressed by existing designs at the same time.

As to the actuation stage, we utilize a piezoelectric (PZT) actuator in this article to ensure a high resolution, fast response speed, and high bandwidth [29], [30]. To address the hysteresis problem of PZT, we apply a sliding mode controller with an uncertainty and disturbance estimator (UDE) (SMC-UDE), as discussed in [31], [32], and [33]. Existing disturbance suppression method, such as the Prandtl–Ishlinskii (PI) model [34], [35] controller, active disturbance rejection control [36], UDE [37], and disturbance-observer-based controller [38] are all potential alternatives, which is not the main focus of this article and will not be specified in detail.

Addressing the aforementioned technical challenges and limitations of the existing methods, the main contribution of this article can be summarized as follows.

- 1) Proposing an RT estimation scheme for vehicle vibration with multiple time-varying dominant frequencies and broad bandwidth.
- 2) The proposed scheme, including the adaptive filters and forecasting LSTM-RNN, is capable of eliminating the phase shift in the estimation.
- 3) Realizing RT active compensation in vibration control simulation for vehicle-based long-distance laser emission tasks.

The rest of this article is organized as follows. Section II introduces the proposed method. In Section III, the experiments and results are presented and discussed. Finally, Section IV concludes this article.

II. METHODOLOGY

As shown in Fig. 1, the vibration θ from various sources leads to a positioning error of the vehicle-based laser. To compensate for it, the proposed RT active compensation method applies gyroscopes and PZT actuators for vibration estimation and compensation actuation, respectively. In the estimation stage, the raw sensing result includes noise, drift derivative, and is delayed. The proposed scheme consisting of the adaptive filter RLS-F and forecasting LSTM-RNN eliminates these disturbances. The low-pass RLS-F first removes the noise n . The signal is then integrated to obtain vibration θ_d and results in drift d . The high-pass RLS-F is then applied to remove drift. The potential phase shift from the conventional filters is hence avoided. Next, the LSTM-RNN performs forecasting to eliminate the delay-induced phase shift and outputs θ_{out} as the estimation of the vibration which is then proceeded to the control algorithm for

PZT. The output actuation $\tilde{\theta}$ is of the opposite direction to θ , and the laser path is hence corrected. The detailed derivation of the proposed RLS-F and LSTM-RNN is as follows.

A. RLS-F

The RLS-F is a type of adaptive filter, which in this article is applied to remove the sensor noise and integration drift. In noise removal, a low-pass RLS-F is applied; while in drift removal, a high-pass RLS-F is applied. Regardless of filter type, the RLS-Fs share the same regression algorithm. The RLS-F models the input signal as a polynomial function of time and step-wisely updates the function parameters. Meanwhile, by introducing a forgetting factor λ , it continuously reduces the effect of historical data points. Therefore, the model convergence rate is more rapid and signal domains with shorter periods (i.e., higher frequency) are eliminated in the model, which is equivalent to filtering. And by selecting either the model or the error term as output, the filter type (either low-pass or high-pass) is specified. Compared to the conventional filters, the phase shift of RLS-F is negligible. Hence, in the proposed estimation scheme, the phase difference is only from the system delay which is a fixed amount. The derivation is as follows.

1) *Derivation:* The RLS-F accepts the input signal and models it as a function of time. Hence, the model y of each time step is

$$y_t = \mathbf{w}_t \mathbf{x}_t. \quad (1)$$

From observations, in this research, the modeling function selection has little effect on the steady-state error. For simplicity, we apply the polynomial function and specify (1), as: $\mathbf{w}_t = [w_0, \dots, w_n]$ includes the parameters, $\mathbf{x}_t = [1, \dots, t^n]^T$ is the reference vector, and n is the order of the modeling function.

The modeling error is then calculated with reference to the input s_t

$$e_t = s_t - y_t. \quad (2)$$

In standard least square algorithm, the cost function to be minimized is defined as

$$C(\mathbf{w}_t) = \sum_{i=0}^t e_i^2. \quad (3)$$

The RLS algorithm further modified the function by introducing a forgetting factor λ typically ranges from 0.8 to 1

$$C(\mathbf{w}_t) = \sum_{i=0}^t \lambda^{t-i} e_i^2. \quad (4)$$

Clearly, older data are gradually weighed less as the forgetting factor is introduced for every time step. Hence, the model y tends to converge to signal domains with less variance in amplitude, i.e., lower frequency. This is equivalent to low-pass filtering. Conversely, the high-frequency domains are filtered out as modeling error e , as their amplitudes variate more frequently and can hardly accumulate effect before being “forgotten.” Selecting either y or e as the output of the RLS-based filter specifies the filter type as low-pass or high-pass accordingly.

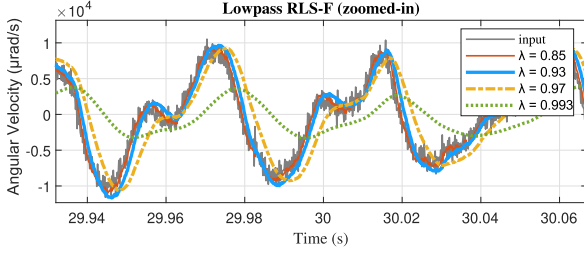


Fig. 2. Lowpass RLS-F to the raw gyroscope input with different λ applied.

And to minimize the cost function and obtain the proper weight vector $\mathbf{w}_t$, the partial derivative is derived and set to 0. Substituting (1) and (4), there is

$$\frac{\partial C(\mathbf{w}_t)}{\partial w_j} = - \sum_{i=0}^t 2\lambda^{t-i} e_i t^{i-j-1} = 0, \text{ where } j \in [0, n]. \quad (5)$$

Substituting (2), and rearrange in matrices format for weight vector $\mathbf{w}_t$ calculation, there is

$$\mathbf{w}_t = \mathbf{R}_t^{-1} \mathbf{V}_t \quad (6)$$

where

$$\mathbf{R}_t = \sum_{i=0}^t \lambda^{t-i} \mathbf{x}_i \mathbf{x}_i^T, \text{ and } \mathbf{V}_t = \sum_{i=0}^t \lambda^{t-i} e_i \mathbf{x}_i. \quad (7)$$

As solving for (6) involves matrix inversion, the RLS algorithm applies a regression manner for ease of implementation

$$\mathbf{R}_t^{-1} = \mathbf{P}_t = \frac{1}{\lambda} (\mathbf{P}_{t-1} - \mathbf{K}_t \mathbf{x}_t^T \mathbf{P}_t) \quad (8)$$

with

$$\mathbf{K}_t = \frac{\mathbf{P}_t \mathbf{x}_t}{\lambda + \mathbf{x}_t^T \mathbf{P}_t \mathbf{x}_t}. \quad (9)$$

And

$$\mathbf{V}_t = \lambda \mathbf{V}_{t-1} + e_t \mathbf{x}_t. \quad (10)$$

Substituting (8) and (10) to (6), the weight vector is hence updated as

$$\mathbf{w}_t = \mathbf{w}_{t-1} + \mathbf{K}_t e_t. \quad (11)$$

2) Forgetting Factor: The selection of λ is also essential to the filter design, which mainly affects the cutoff frequency f_c . A typical λ ranges from 0.8–1. Regardless of the filter type, a larger λ refers to lower f_c . To better illustrate this, in Fig. 2, we take the low-pass RLS-F for example, where y is selected as output. Clearly, as λ increases, less high-frequency domains are kept, indicating lower f_c . However, λ selection also affects modeling accuracy. The determination of the λ value is based on both effects to obtain the right f_c and high accuracy.

Besides, unlike conventional filters, the RLS-F is based on the model convergence in the time domain, irrelevant to the frequency domain. And the model convergence is also dependent on the input amplitude. Hence, for signals with different amplitudes, though with the same λ , f_c would vary. However,

for each sensor, there is no necessity to change the λ , as noise and drift are intrinsic to sensors. In this article, we searched through the possible range and determine the λ for the low pass to be 0.93, for the high-pass filter to be 0.993.

B. LSTM-RNN

As shown in Fig. 1, the LSTM-RNN is proposed to eliminate the delay-induced phase shift in θ_d by forecasting the future value. Accepting idea of traditional forecasting techniques, the network takes a set of historical data as input and is trained to output a future data point of a fixed time interval. The constant system delay and the resultant phase shift is hence eliminated. The proposed network consists of an input layer, an LSTM layer, three fully connected layers, and an output layer, as shown in Fig. 3. A sliding window of size M is applied to form the input vector. The LSTM layer possesses states of size N . Three fully connected layers each possess output size of F_1 , F_2 , and 1. The layers are activated with the rectified linear unit (ReLU) function. The design takes advantage of LSTM layer in sequence processing while maintaining simplicity for RT application.

1) Parameters Selection: To include necessary information for forecasting, a larger M is desired. However, the vibration features of vehicles are inconsistent. A long input vector may affect the forecasting of the most recent variance of the input. Furthermore, the RT implementation also limits the network complexity. A large M significantly increases computation needed. Therefore, in consideration of all the factors, M is set to be 20 in this article.

Similarly, the state vectors sizes N of the LSTM layer and the sizes of the fully connected layers which increase the depth of the network and enable better state representation and generalizability, are also important to the forecasting performance. However, at a cost of computation increment, the RT implementation is affected, and the training data demand is higher. To determine the network structure, we begin with a simple design and increase the complexity gradually as needed and as the hardware allows. And we finalize the current structure as $N = 4$, $F_1 = 10$, and $F_2 = 2$, and the ReLU function [39] is applied for activation to perform nonlinear mapping. The proposed structure well balances the computation simplicity for RT implementation and accuracy in forecasting.

2) LSTM Unit: The LSTM is an advanced structure of RNN, which is developed to solve the vanishing gradient problem [40]. The design is proved to possess better performance for long time series processing than classic RNN with data efficiency in modeling compared with Transformer [41]. A typical LSTM unit possesses a forget gate, an update gate, and an output gate. Two state vectors, namely, the cell and the hidden states are updated for each time step. In this article, we accept the classic LSTM unit with a forget gate [42], as shown in Fig. 2.

For each time step, the LSTM unit accepts the input time series $\mathbf{X}_t$ and states from the last time step. The input gate $\mathbf{i}_t$ controls the information accepted and preserved in the cell $\mathbf{c}$; the forget gate $\mathbf{f}_t$ takes the contrary effect to reset partial information. A cell candidate is calculated by $\mathbf{g}_t$, based on which the latest cell state $\mathbf{c}_t$ and hidden state $\mathbf{h}_t$ are updated. The detailed derivation

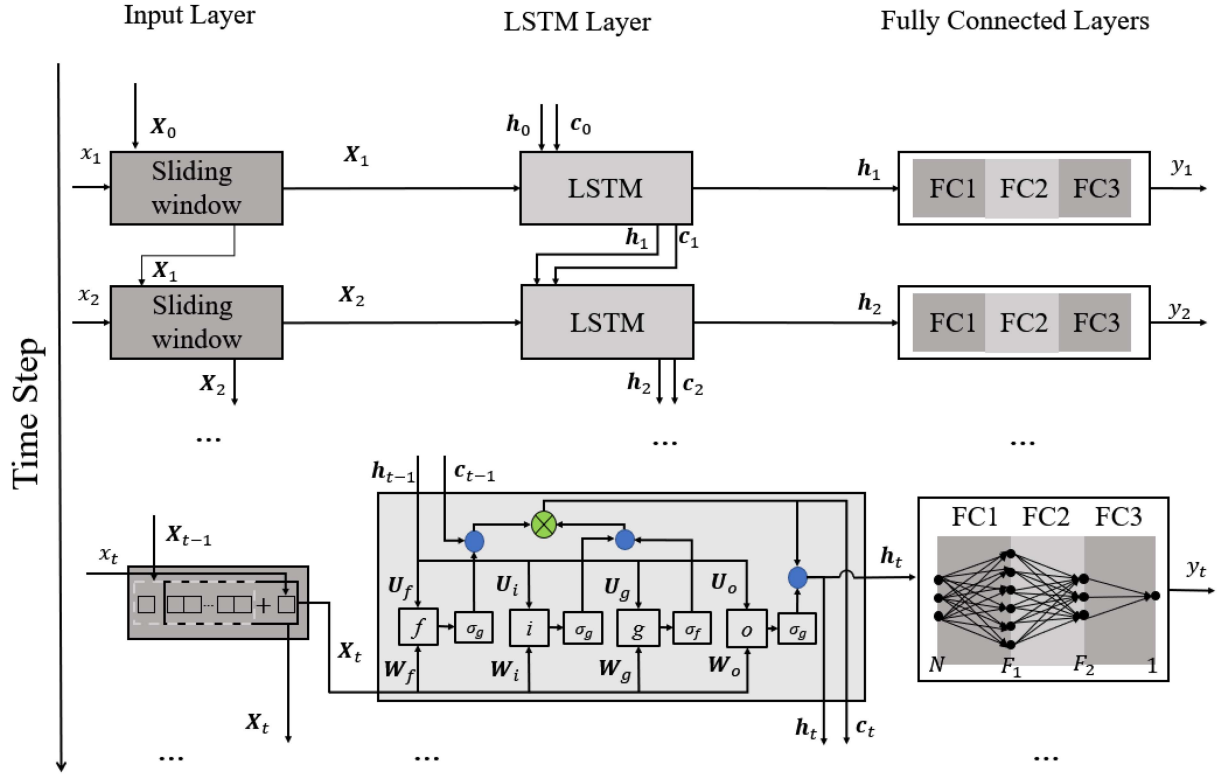


Fig. 3. Proposed LSTM-RNN forecasting network. For each time step, the network accepts the input vector and states from the last step, and outputs the forecast result and optimized state vectors. In LSTM unit, the blue dot refers to the elementwise product; while the green dot refers to summation operations.

is as follows:

$$\mathbf{i}_t = \sigma_g(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i) \quad (12)$$

$$\mathbf{f}_t = \sigma_g(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f) \quad (13)$$

$$\mathbf{o}_t = \sigma_g(\mathbf{W}_o \mathbf{x}_t + \mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{b}_o) \quad (14)$$

$$\tilde{\mathbf{c}}_t = \sigma_f(\mathbf{W}_g \mathbf{x}_t + \mathbf{U}_g \mathbf{h}_{t-1} + \mathbf{b}_g) \quad (15)$$

$$\mathbf{c}_t = \mathbf{f}_t \circ \mathbf{c}_{t-1} + \mathbf{i}_t \circ \tilde{\mathbf{c}}_t \quad (16)$$

$$\mathbf{h}_t = \mathbf{o}_t \circ \mathbf{c}_t \quad (17)$$

where the function $\sigma_g(\cdot)$ is the activation function of the gates. In this article, Sigmoid function is used, as introduced in [43]. Similarly, $\sigma_f(\cdot)$ is the output activation function, where the hyperbolic function is used.

The input vector $\mathbf{X}_t$ is of size $M \times 1$. In this article, it consists of the most recent M data points of the vibration signal. The cell state $\mathbf{c}$ and the hidden state $\mathbf{h}$ of every time step is a vector of size $N \times 1$.

$\mathbf{W}_i, \mathbf{W}_f, \mathbf{W}_o$, and $\mathbf{W}_g$ are the weight vectors of size $N \times M$; $\mathbf{U}_i, \mathbf{U}_f, \mathbf{U}_o$, and $\mathbf{U}_g$ are the recurrent weight vectors of size $N \times N$; $\mathbf{b}_i, \mathbf{b}_f, \mathbf{b}_o$, and $\mathbf{b}_g$ are the bias vectors, of the same size as the state vectors.

3) Training Approach: The training approach is performed offline, before online implementation. To train the network, the lost function is defined as the half mean square error of the

network response with respect to the targeting output, as

$$L = \frac{(\theta_t - \tilde{\theta}_t)^2}{2} \quad (18)$$

where θ_t and $\tilde{\theta}_t$ refer to the ground truth and the network response, respectively. In this article, we apply the future value in a number of time steps ahead of the current time step as the ground truth for forecasting network training. In practice, the forecasting interval is dependent on the amount of intrinsic sensing delay. From our study, this delay is mostly constant and possesses negligible frequency dependency. Hence, a constant forecasting interval is applied. For a system, the delay can be determined through a one-time calibration by inputting a standard signal to the system and collecting the sensing result for comparison. A slightly larger forecasting interval than the delay is recommended to leave room for future adjustment, however, a too wide interval may affect the forecasting accuracy. In this article, the delay is around 25 time steps (5 ms), and a forecasting interval of 30 steps with an additional delay of 5 time steps is applied.

We experiment with different optimizers (e.g., SGD, Adam, RMSprop, Adagrad) to find the best option. Ultimately, we use an Adam optimizer, as discussed in [44] and [45], with decay factors $\beta_1 = 0.9$ and $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. We test various learning rates (e.g., 0.001, 0.01, 0.1) to find the one that allows stable convergence and reasonable training time. In the final version, we keep the learning rate constant at $\alpha = 10^{-3}$.

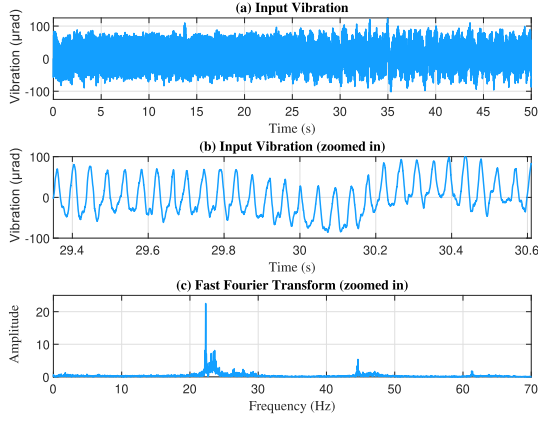


Fig. 4. One of the vibration data sets sampled from the truck and its FFT plot. As shown, there are multiple dominant frequencies up to around 60 Hz.

In addition, we experiment with different batch sizes to balance training speed and convergence quality, ultimately using a mini-batch size of 1600 for each training update. We also add dropout and/or regularization (e.g., L1 and L2) to avoid overfitting and test different rates and strengths to find the best fit. In the final version, a dropout rate of 0.15 is applied. In addition, we set the maximum number of epochs to 10^4 and shuffle the training sets for every epoch, further optimizing the model's performance. The training process will stop as error performance index is satisfied or the maximum iteration is reached. In practice, we have observed that the current maximum iteration can converge in most cases, providing satisfactory results for various applications.

III. EXPERIMENT AND RESULT

To validate the performance of the proposed method, we conducted three simulations and one online RT experiment, in which the proposed approach is compared with existing methods. In these experiments, vibration data sampled from a truck-based (Ford Motor Comp.) laser emission system is applied, as shown in Fig. 1. The signal and fast Fourier transform (FFT) plot are shown in Fig. 4. Clearly, there are multiple dominant frequencies in bandwidth up to 60 Hz. The root mean square error (RMSE) with respect to the ground truth of each experiment is calculated for evaluation.

A. Simulation

1) *RLS-F Validation*: The RLS-F is proposed for denoise and drift removal. The performance is compared with the conventional Butterworth filter in both low-pass and high-pass filtering. The filtfilt zero-phase filter is applied as ground truth for evaluation purposes. The parameters used in the experiment are as follows. The Butterworth filters take second order, and cut off at 100 and 2 Hz, respectively. The RLS-F applies $\mathbf{P}_0 = 10^6 \times \mathbf{I}$, $\mathbf{w}_0 = \mathbf{0}$, λ at 0.93 and 0.993, respectively.

The initialization of the RLS-F follows the discussion in [46], that the initial $\mathbf{P}$ is set as $\delta^{-1} \times \mathbf{I}$, where δ is a small positive constant, and the initial weights are all set to 0. Alternatively,

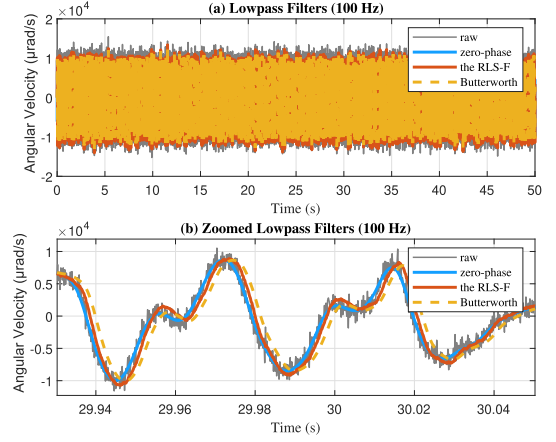


Fig. 5. Comparison of low-pass filters' result. The sensor noise in raw data is removed by both methods, but the proposed RLS-based filter possesses less phase shift.

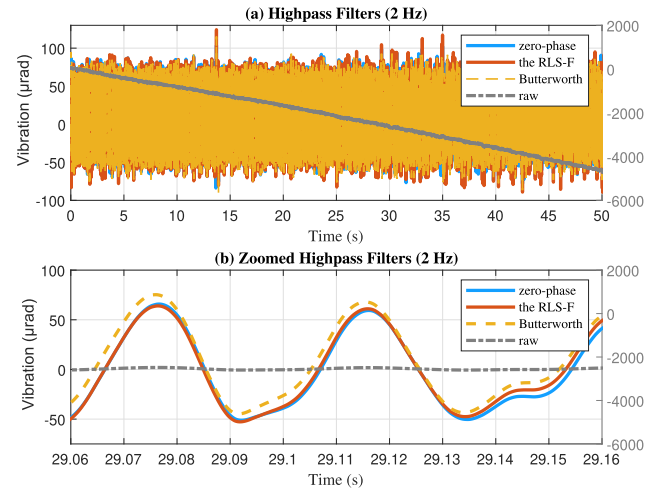


Fig. 6. Comparison of high-pass filters' result. The drift is removed in both filtering results, however, the proposed RLS-based filter avoid most phase shift in filtering. The raw input follows the right axis.

based on (8) and (10), the initial values can be calculated. However, the calculation requires the inversion of matrix and is input-dependent. For ease of implementation, we select the first way and tried with different constants and set $\delta^{-1} = 10^6$ for this article.

The low-pass filter comparison is shown in Fig. 5. Clearly, the RLS-F possesses significantly less phase shift than the conventional filter. Though the results are visually similar, the RLS-F possesses much higher accuracy and less error. The RMSE of RLS-F is $1.341 \times 10^3 \mu\text{rad/s}$, which is 24.13% of the zero-phase rms; while the value for the Butterworth filter is $2.338 \times 10^3 \mu\text{rad/s}$ and 42.07%.

Similarly, the high-pass filtering comparison for drift removal is shown in Fig. 6. The RMSE of the RLS-F is $7.811 \mu\text{rad}$, 19.17% of the zero-phase rms; while the Butterworth filter possesses $8.920 \mu\text{rad}$ RMSE, 21.81% of the ground truth rms. The proposed RLS-F also has less phase shift and higher accuracy.

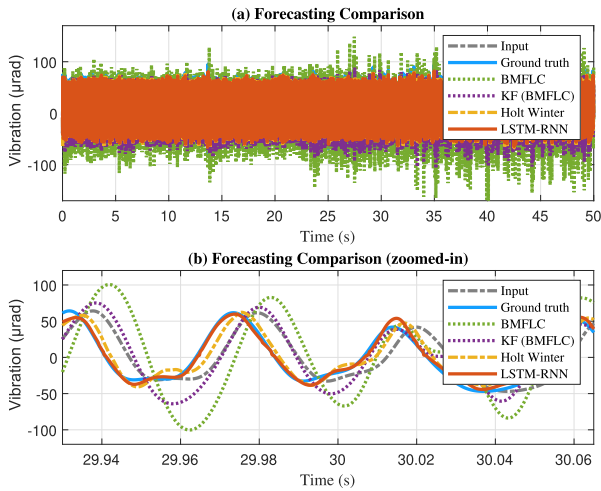


Fig. 7. Comparison of the time series forecasting methods. The proposed method possesses higher accuracy which is close to the ground truth.

From the figures, the multiple time-varying dominant frequencies and broad bandwidth of the signal being processed by the RLS-Fs are well preserved. Moreover, in the discussed RT compensation, the phase shift of conventional filters is a critical problem not only affecting the filtering but also degrading the accuracy of the forecasting. As introduced, there is already a phase shift caused by delay to be eliminated. An additional frequency-dependent phase shift can hardly be specified and further removed, which greatly degrades the filtering accuracy and estimation performance. Conversely, the proposed RLS-F is based on the adaptive RLS algorithm, which possesses little phase shift. The phase difference between the estimated signal and the actual vibration is only from the constant system delay.

2) LSTM-RNN Validation: In this simulation, the forecasting performance of the proposed LSTM-RNN is compared with the Holt Winter's method, FLC-based algorithm, and the FLC-based predictive Kalman filter (KF). All methods forecast the value of a vibration signal with sensor noise and drift removed by zero-phase filter to avoid disturbance. The forecasting interval is 6 ms (30 time steps).

The Holt-Winter's method applies a seasonal factor of 80, meaning at least 80 data points to be stored in the memory, which is challenging to most RT computers. However, it is necessary to avoid divergence. And the α , β , and γ are set as 0.8, 0.2, and 0.005, respectively.

As to the FLC methods, we apply a band-limited multiple fourier liner combiner (BMFLC) following discussion in [8], as there are multiple dominant frequencies. The method is usually used as a single-step method, but in this simulation, we only apply it for phase correction. The BMFLC band is set from 2 to 60 Hz, and ten sets of sinusoidal functions are used.

The BMFLC-based phase correction can go one step further, where a predictive KF is applied. The method first applies the BMFLC model for prediction and updates the prediction based on the true value obtained later.

As shown in Fig. 7 and Table I, the proposed LSTM-RNN possesses the best forecasting accuracy and is close to the ground

TABLE I
FORECASTING COMPARISON

	RMSE (μrad)	Percentage (%)
BMFLC	64.93	178.95
KF (BMFLC)	39.12	107.83
Holt Winter's method	12.30	33.91
LSTM-RNN	5.08	13.99
Ground truth	36.28	-

Notes. This table summarizes the forecasting comparison of the proposed method versus other existing methods. The bold entities refers to the least RMSE and RMSE percentage in term of the ground truth RMS.

truth. The RMSE is around $5\mu\text{rad}$, only 14 % of the signal rms. As to the Holt-Winter's method, the RMSE is doubled, because it is less sensitive to the variance of the signal, and has high data demand. In the RT application, keeping the latest 80 data points is necessary to avoid divergence, taking much RAM storage. The BMFLC model is difficult to converge as the dominant frequencies are not consistent. Though with the phase correction term, there is still a significant phase error, causing the forecasting RMSE to even exceed the ground truth rms. With the additional KF, the performance is improved but is still 7.83 % more than the ground truth rms. The proposed LSTM-RNN takes advantage of the LSTM unit's high accuracy in sequence processing. The multiple dominant frequencies and broad bandwidth features of the vibration signal are well recognized and the time variance of the signal is well captured as the state vectors are updated step-wisely. In addition, compared to the Holt-Winter's method, the LSTM-RNN requires much less data to be processed at a time.

Moreover, compared to the FLC and predictive filter, the LSTM-RNN avoids constructing an analytical model, which is challenging due to the complexity of the vibration features. From the results, the BMFLC model is not reliable, degrading the forecasting accuracy badly. The high forecasting accuracy of the proposed LSTM-RNN for signals with multiple dominant frequencies and broad bandwidth, is one of the main contributions of this article.

3) Complete Estimation Scheme: In this simulation, the effectiveness of the proposed multistep scheme (as shown in Fig. 1) for vibration estimation is validated by comparing it with the single-step BMFLC method, the BMFLC-based KF, and a multistep approach which consists of conventional filters and the Holt-Winter's method.

The simulation is conducted based on a PZT fast steering mirror (FSM) laser path to imitate the vehicle-based laser emission encountering vibration. The setup is based on a breadboard, as shown in Fig. 8. A laser (VLM2 635 mm, Coherent) is pointed through the FSM, two beam splitters (50-50, Thorlabs), and to a Position Sensitive Device (PSD, xy4QD, MRC). The FSM is actuated by the PZT actuator (S340, PI). The vibration is sensed by a gyroscope mounted on the mirror. An RT computer (CRIO-9039, NI) equipped with field-programmable gate array (FPGA) module collects and processes the signal to perform the compensation. A camera (acA1440-73gm, Basler) and a standard optical target (R3L3S1N, Thorlabs) are used as visual assistant.

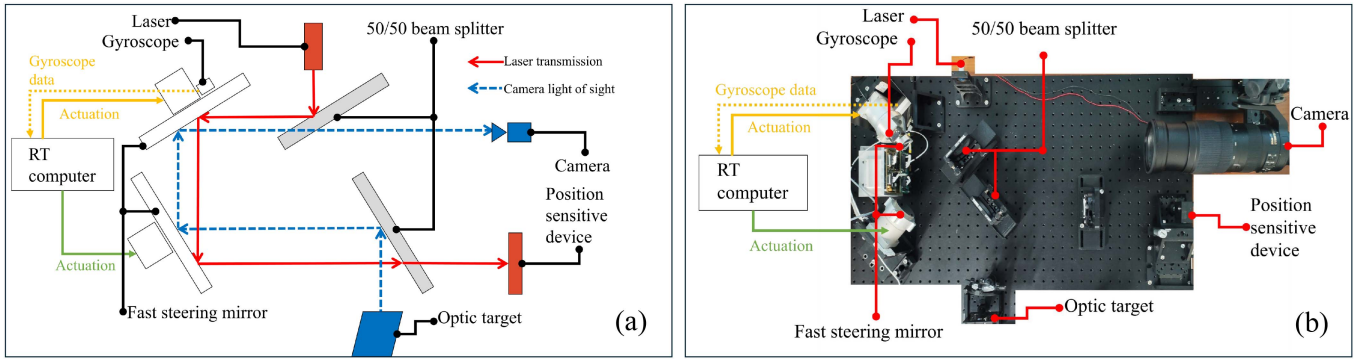


Fig. 8. Laboratory setup. The laser path and related elements are notified in red; while the camera path is in green. (a) Setup and optical paths. (b) Picture of the setup.

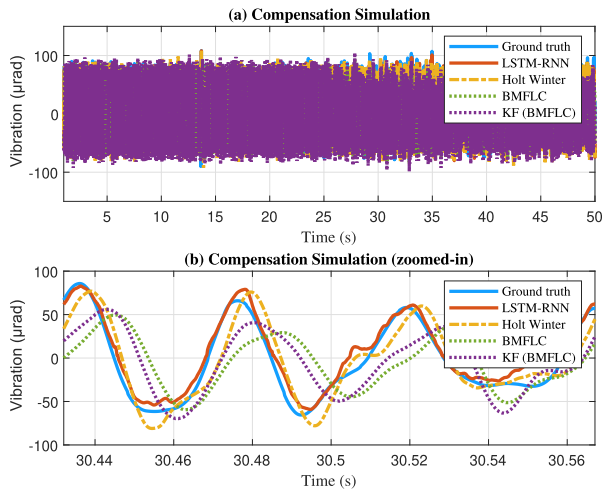


Fig. 9. Simulation comparison of the proposed approach and the existing methods. As shown, the proposed method possesses the least phase shift and is the closest to the ground truth.

TABLE II
COMPENSATION SIMULATION RESULTS

	RMSE (μrad)	Percentage (%)
BMFLC	39.66	95.34
KF (BMFLC)	33.21	79.81
Holt Winter's method	19.52	46.92
LSTM-RNN	9.57	23.00
Ground truth	41.60	-

Notes. This table summarizes the compensation simulation results of the proposed method versus existing methods. The bold entities refers to the least RMSE and RMSE percentage in term of the ground truth RMS.

In this simulation, the vibration signal is introduced to the laser path by FSM and is sensed by the gyroscope. These steps are conducted in RT and the collected raw gyroscope signal is processed with the proposed approach and existing methods to estimate the input vibration in offline. The estimation accuracy of the methods is compared to validate the effectiveness.

The simulation result is shown in Fig. 9 and Table II. Clearly, the proposed multistep approach possesses the highest accuracy.

The RMSE of the proposed estimation scheme is $9.57 \mu\text{rad}$, 23% of the ground truth rms. The result is consistent with the results in simulations 1 and 2. The proposed method well estimates the vibration signal with multiple time-varying dominant frequencies and broad bandwidth, validating the contribution of this research. The existing methods are significantly affected by the inherent sensor delay and the conventional filter phase shift. The BMFLC and the predictive filter can hardly make the model converge before the signal features start to vary. The RMSE of the two methods are all excess 100% of the ground truth rms, making them not suitable for the compensation works. As to the Holt-Winter's method, the RMSE is about double of the proposed scheme result. In rms, the error is about 50% of the input, indicating the estimation is inaccurate.

Compared to the existing methods, the proposed scheme applies the adaptive RLS-F, which avoids additional frequency-dependent phase shift. Compared to the forecasting simulation, the performance of the Holt-Winter's method is degraded as the conventional filters are applied. In practice, we increase the number of steps to be forecast, however, the additional phase shift is frequency-dependent and hard to be specified. A larger interval also degrades forecasting accuracy. As a result, the RMSE is much larger than the proposed method.

As to the BMFLC, the method takes a single-step approach where the gyroscope input is modeled to remove sensor noise, and then perform integration and phase correction using the model. The predictive filter also applies this model for forecasting. Hence, the model converges is essential. As shown, this is challenging due to the dominant frequency, broad bandwidth, and the variance in the feature of the vibration signal. The model convergence is slow and causes additional delay. In practice, a larger forecasting interval is applied. However, as stated, a larger forecasting interval addresses the variance in signal features worse. The accuracy is hence affected.

Conversely, the proposed LSTM-RNN is model-free, avoiding this limitation. Moreover, the proposed LSTM-RNN is a dynamic predicting network, unlike the traditional NN, it is capable of keeping the stored information up to the current time step by updating the states vectors. Therefore, it possesses improved overall accuracy and good generality as the vibration

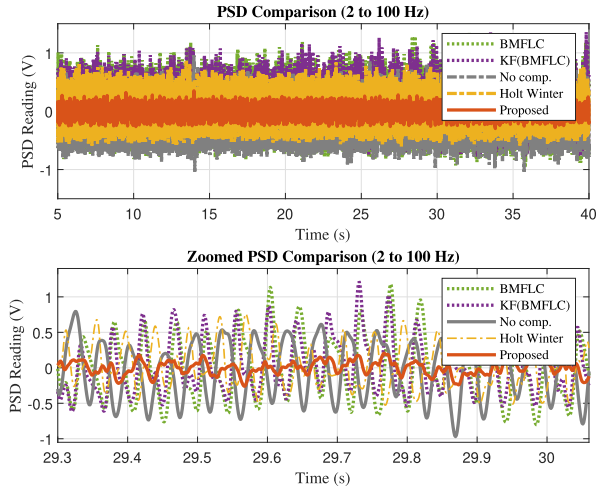


Fig. 10. Online experiment results. A PSD sensor is applied to measure the stability of the laser path after performing vibration compensation. A smaller PSD reading refers to better performance.

TABLE III
ONLINE EXPERIMENT RESULTS

	RMS (V)	Percentage (%)
BMFLC	0.4424	110.99
KF (BMFLC)	0.3911	98.12
Holt Winter's method	0.3047	76.44
LSTM-RNN	0.1005	25.22
Control group	0.3986	-

Notes. This table summarizes the compensation online experiment results of the proposed and existing methods. The bold entities refers to the least RMSE and RMSE percentage in term of the ground truth RMS.

features vary over time. The RLS-F also avoids most additional phase shift as the conventional filters do, making the network forecast only a constant delay. The estimation accuracy is hence improved. The simulation results validate the effectiveness of the proposed techniques and the contribution of this work.

B. Online Experiment

The online experiment means to validate the RT performance of the proposed approach, as the proposed estimation and compensation methods typically require highly on computation, making RT application the main difficulty in use. The setup is as shown in Fig. 8, however, unlike the simulation, as one mirror of the FSM introduces the vibration, another mirror actively compensates it following the proposed method running in RT on the FPGA module of the computer. As the second mirror is rotated with the same amplitude but opposite direction to the first mirror, as long as the proposed approach is effective, a stable laser path will be observed. The PSD is used to measure the stability of the laser spot for evaluation.

The compensation result is shown in Fig. 10 and Table III. The proposed method compensated most of the vibration and maintains a stable laser path. The PSD reading rms of the proposed compensation is 0.1005 V which is 25.22 % of the control group result, indicating a compensation rate of 74.78 %, which is

consistent with the simulation result shown in Table II, indicating the proposed estimation is capable for RT application, and the proposed compensation is reliable, which is a key contribution of this research.

The online experiment also validates the computational efficiency of the proposed LSTM-RNN for RT usage. Traditional LSTM-RNNs typically possess great complexity and high data demands. In this article, the proposed design is specifically based on the identified technical challenges to balance simplicity and performance with high accuracy. The vibration signal features, i.e., the multiple time-varying dominant frequencies and broad bandwidth, have not been discussed much in existing works.

As to the existing methods, the compensation is not successful. The residue vibration of BMFLC-based compensation are both about 100 % of the control group. The Holt-Winter's method is a bit better, but there is still more than 75 % vibration not compensated. The movement of the laser spot is very obvious which would lead to positioning failure if applied in the real world. Clearly, due to the phase shift in conventional filters and the modeling inaccuracy, though the estimation is visually similar, as shown in Fig. 9, the actual error in compensation is significant.

Besides, in online experiments, there are factors that may amplify this error, making the results not very consistent with the simulation results in Table II. For FSM system, if the compensation is not accurate, the residue rotatory vibration will also induce translation movement of the laser path, and result in additional positioning error. Besides, to ensure RT application, the FPGA platform is applied in online experiments, on which the calculation is run in fixed-point format. Unlike PC platforms, the precision of the numbers is fixed. As the algorithms run, there is more or less rounding of the intermediate values. The accuracy may hence be affected.

IV. CONCLUSION

In this article, an RT vibration estimation and compensation scheme for vehicle-based high-precision devices is proposed and validated. The vibration typically possesses multiple time-varying dominant frequencies and broad bandwidth. There is also an inherent delay-induced phase shift. The proposed method well addresses these challenges and realizes 75% compensation rate in RT active operation.

For future work, field experiments for a real-world vehicle-based laser stabilization for communication will be considered, as the current setup is laboratory-based. Furthermore, the current setup includes only a gyroscope as the sensor. Additional sensors like the camera will be investigated to improve the accuracy by sensor fusion.

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